PROJECT CHARTER DOCUMENT

Project Name: Sprintly (Customizable Kanban Workflow System)
Version: 1.1
Date: 04/03/2026
Project Duration: 30 Days
Prepared By: Team Blitzkrieg

Purpose

To develop a web-based Kanban project management system with customizable workflows that enables teams to collaboratively manage tasks, enforce structured role-based control, and provide transparent project visibility through a simple and lightweight interface.

Objective

- Enable users to register, log in, and create teams and boards.
- Allow Owners to define and customize workflow stages (e.g., To Do, In Progress, Done, etc.).
- Implement role-based access control (Owner, Participant, Viewer).
- Ensure every task has mandatory ownership assignment.
- Enable drag-and-drop task movement with permission-based restrictions.
- Provide dashboard insights including task distribution, completion percentage, and overdue tracking.
- Allow view-only access for stakeholders without modification privileges.

Demand

This project addresses the need for a structured yet lightweight task management system tailored for small teams, academic groups, and early-stage startups.

Many existing tools are either too complex for small teams or lack strict workflow and role enforcement. This solution fills the gap by offering customizable workflows with controlled permissions, improving accountability, transparency, and overall project efficiency without enterprise-level complexity.

Business Requirement

● Three Project-Level Roles: Owner, Participant, Viewer

Owner:

- Create and manage teams and boards
- Define and customize workflow stages
- Create, assign, edit, and delete tasks
- Move any task across stages
- Invite and manage team members
- View dashboard analytics and project insights

Participant:

- View project board and tasks
- Move only tasks assigned to them
- Comment on tasks
- View dashboard metrics

Viewer:

- View boards, tasks, and workflow stages
- Monitor progress and dashboard insights
- No modification permissions (read-only access)

Dashboard & Progress Monitoring

• Display number of tasks per workflow stage
• Display project completion percentage
• Highlight overdue tasks
• Provide high-level project visibility for monitoring stakeholders

Team Invitation & Access Control

• Owners can invite users via email or system-based invitation
• Role assignment is defined at the project level
• Role-based access control is strictly enforced across UI and APIs

Technical Requirement

User Authentication & Authorization

- UI and API routes for:
  - Registration
  - Login
  - Logout
    - JWT-based authentication
    - Role-based access control middleware
    - Secure session handling

Project & Role Management

- API endpoints to:
  - Create teams
  - Create boards/projects
  - Invite users
  - Assign roles (Owner, Participant, Viewer)
    - Backend validation to enforce permission boundaries

Workflow Management

- API to create, update, reorder, and delete workflow stages
- Validation to prevent deletion of stages containing active tasks (until removed)
- Dynamic workflow rendering on frontend

Task Management

- CRUD APIs for task creation and modification
- Drag-and-drop integration on frontend (React-based)
- Backend validation for task movement permissions
- Status change logging mechanism
- Overdue task detection logic

Dashboard & Reporting

- API to fetch:
  - Task count per stage
  - Completion percentage
  - Overdue tasks list
    - Dashboard UI for visual representation of project health

Technological Requirement

Frontend

- React Js
- Tailwind CSS

Backend

- Java Spring Boot

- RESTful APIs
- JWT Authentication

Database
- PostgreSQL

Tools
- Visual Studio Code / IntelliJ IDEA
- GitHub (Version Control)
- Postman (API Testing)

Stakeholders

Stakeholder	Role	Responsibility
Project Development Team	Implementation	Design, development, testing
Academic Supervisor / Evaluator	Oversight	Review & evaluation
Product Owner	Requirement Authority	Requirement validation
Owners (End Users)	Primary User	Manage projects and tasks
Participants	Active User	Execute assigned tasks
Viewers	Monitoring User	Monitor progress

Project Organization

Team Structure

Role	Count	Responsibility	Assigned Members
Frontend Developer	3	UI development, board design, drag-and-drop implementation, dashboard UI	Shreyash Kumbhar, Ayush Chandekar,PrachiPhunde
Backend Developer	3	API development, business logic implementation, authentication, API testing	Shreyash Kumbhar, Ayush Chandekar,Yashvardhan Jangid
Database Designer	2	Database schema design, relationship modeling, optimization, transactional integrity	Prachi Phunde, Ayush Chandekar
Project Management	2	Planning, sprint tracking, milestone monitoring, risk management, coordination	Prachi Phunde, Ayush Chandekar
Documentation Team	2	SRS, BRD, Charter preparation, technical documentation	Shreyash Kumbhar, Yashvardhan Jangid

Tester / QA	4	Functional testing, API testing, integration testing, bug reporting	Internal : Yashvardhan Jangid Ayush Chandekar External : Sarvesh Pandey, Pratik Bhosale

PESTEL Analysis

- **Political:**
Minimal political impact. However, compliance with national IT policies and data protection regulations must be considered if deployed publicly.

- **Economic:**
Provides a low-cost productivity solution for small teams, startups, and academic groups who cannot afford expensive enterprise tools.

- **Social:**
Supports remote collaboration and structured teamwork, improving accountability and transparency in academic and professional environments.

- **Technological:**
Leverages modern web technologies (React, Spring Boot, REST APIs, JWT) to ensure scalability, modularity, and maintainability.

- **Legal:**
Must comply with data privacy and security standards, including secure password storage, HTTPS communication, and proper access control enforcement.

- **Environmental:**
Reduces dependency on physical meetings and paperwork by enabling digital task management and remote collaboration.

Risk Analysis

Risk	Impact	Description	Mitigation
Role-Based Access Control Failure	High	Incorrect permission enforcement may allow unauthorized task edits or data exposure.	Implement strict backend validation and thorough RBAC testing before deployment.
Timeline Overrun	High	30-day deadline may be missed due to integration complexity.	Weekly milestones, parallel frontend-backend development, daily progress tracking.

Drag-and-Drop Performance Issues	Medium	Board UI may lag with large numbers of tasks.	Use optimized React drag-and-drop library and implement pagination or lazy loading if required.
Data Loss / Inconsistent Transactions	High	Improper database handling may cause task or workflow corruption.	Use transactional integrity in backend and regular database backups.
Server Downtime	Medium	Hosting issues may make system temporarily unavailable.	Choose reliable hosting provider with minimum 99% uptime SLA.
Security Vulnerabilities	High	Weak authentication or improper JWT handling could lead to data breaches.	Implement secure password hashing, token validation, HTTPS enforcement, and security testing.
Low User Adoption	Medium	Users may not consistently update tasks, reducing system effectiveness.	Keep UI intuitive and lightweight; provide simple onboarding guidance.
Budget Overrun	Low	Unexpected hosting or infrastructure costs.	Use shared hosting initially and open-source technologies.

Milestones & Timeline (30 Days Plan)

Week	Phase	Key Tasks
Week 1	Requirement Finalization & Design	• Finalize SRS & database schema • Define API contracts • Authentication module design
Week 2	Core Backend Development	• User authentication APIs • Project & role management APIs • Workflow configuration APIs • Task CRUD APIs • Initial API testing
Week 3	Frontend Development & Integration	• Authentication UI • Board UI with drag-drop • Dashboard implementation • API integration • Role-based UI restrictions
Week 4	Testing, Optimization & Deployment	• Integration testing • Performance optimization

		• Bug fixing • Deployment setup • Final review & documentation

Budget Estimate

Item	Estimated Cost (INR)
Hosting (cPanel)	₹10,000–₹15,000/year
Domain	₹500–₹1,000/year
Total Estimated	₹15,000–₹20,000/year